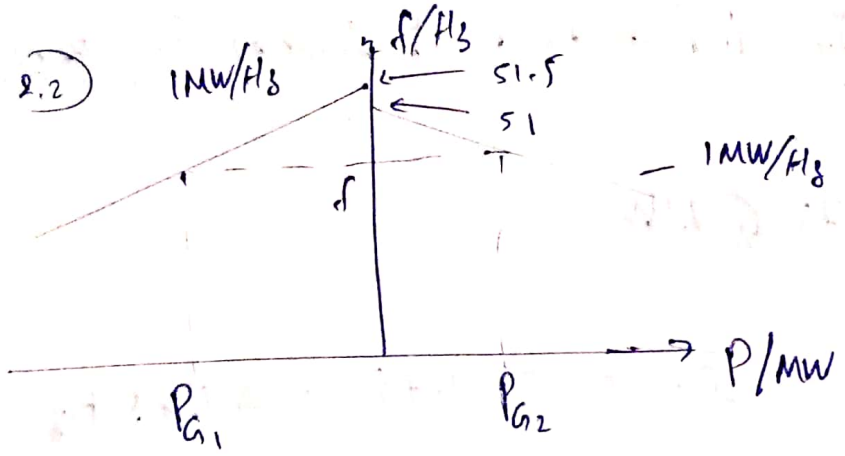
2016

Q2 2.2



(a)

$$1 = \frac{P_{G_1}}{51.5 - f} \rightarrow f = 51.5 - P_{G_1}$$

$$1 = \frac{P_{G_2}}{51 - f} \rightarrow f = 51 - P_{G_2} = 51 - 2.5 + P_{G_1}$$

$$P_{G_1} + P_{G_2} = 2.5 \text{ MW}$$

$$51.5 - P_{G_1} = 51 - 2.5 + P_{G_1}$$

$$2P_{G_1} = 3$$

$$P_{G_1} = 1.5 \text{ MW}, \quad P_{G_2} = 1 \text{ MW}, \quad f = 50 \text{ Hz}$$

(b)

$$f = 51.5 - P_{G_1}$$

$$f = 51 - P_{G_2}$$

$$P_{G_1} + P_{G_2} = 3.5$$

$$51.5 - P_{G_1} = 51 - 3.5 + P_{G_1}$$

$$2P_{G_1} = \cancel{3.5}\ 4$$

$$P_{G_1} = \cancel{1.75}\ 2 \text{ MW}, \quad P_{G_2} = \cancel{1.25}\ 1.5 \text{ MW}, \quad \cancel{f = 49.75 \text{ Hz}}.\ f = 49.5 \text{ Hz}.$$

c) if gen 2 frequency increased by 0.5 Hz.
Power is symmetrically distributed between 2 gens

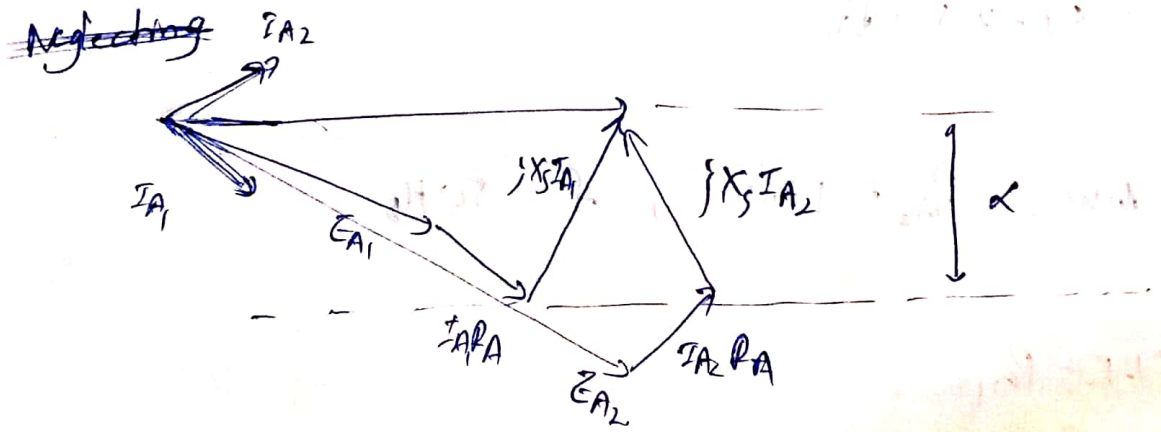
$$P_{n_1} = P_{n_2} = \frac{3.5}{2} = 1.75 \text{ MW}$$

$$\frac{1.75}{51.5 - f} = 1 \rightarrow f = 51.5 - 1.75 = 49.75 \text{ Hz}$$

Q3

3.1) $V_\phi = E_A + I_A R_A + jX_s I_A$

since load is constant, Power is constant.
When I_f increases, as ϕ varies (increases), E_A will increase



as E_A increases power factor goes from lagging to unity to leading

3.2

load 1 : $P = 100\,kW$, $Q = P\tan\theta = 80.23$ kVar

load 2 : $P = 200\,kW$, $Q = P\tan\theta = 150$ kVar

(a) load 3: $P = 150\,kW$, $Q = P\tan\theta = 72.65$ kVar.

$P_{net} = 450\,kW$, $Q_{net} = 302.88$ kVar

~~$S = \sqrt{P_{net}^2 + Q_{net}^2} = 542.44$ 54~~

$S = P + jQ = 541.94 \angle 33.87^\circ$

$S = \sqrt{3}\, V_L I_L$

$I_L = \dfrac{(541.94 \angle 33.87^\circ)\,kVA}{(\sqrt{3} \times 415 \angle 0^\circ)\,V} = 0.75 \angle 33.87^\circ \times 10^3$

$= 750 \angle 33.87^\circ$ A

(b) load 3: $P = 150\,kW$, $Q = -P\tan\theta = -72.65$ kVar

$P_{net} = 450\,kW$, $Q_{net} = 157.58$ kVar.

$S = P + jQ = 476.79 \angle 19.3^\circ$ kVA

$I_L = \dfrac{S}{\sqrt{3}\, V_L} = \dfrac{476.79 \angle 19.3^\circ\,kVA}{\sqrt{3} \times 415\,V} = 663.3 \angle 19.3^\circ$ A

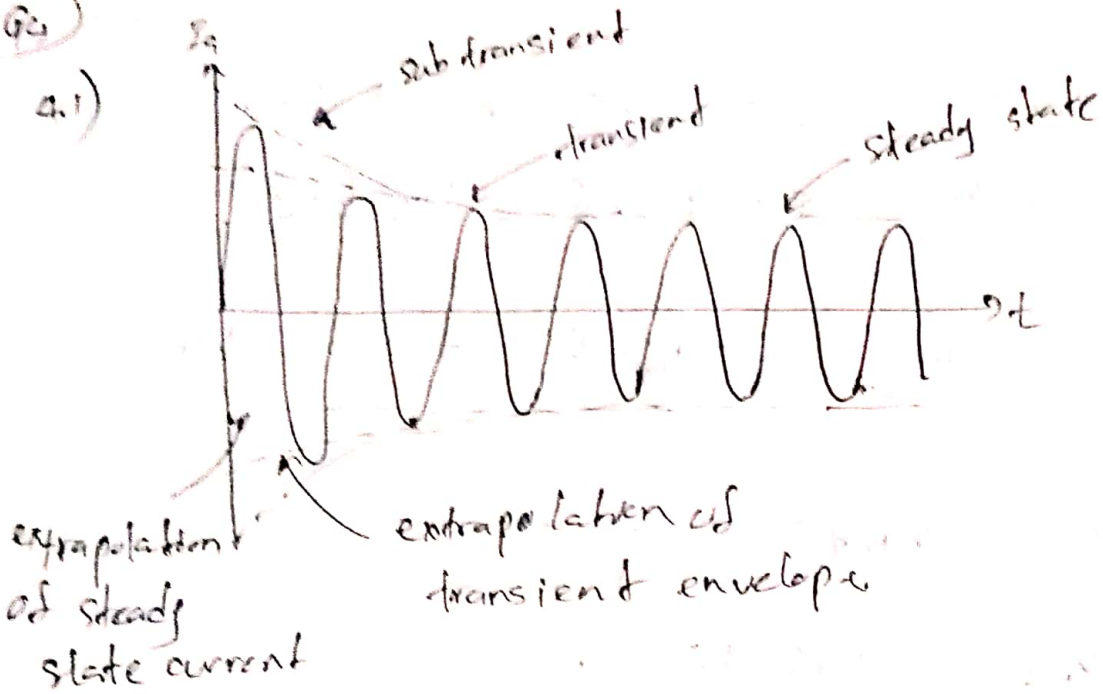
(c) ~~Loss~~ $P_{Loss,A} = 1.6875 \times 10^6 R_L$ W

$P_{Loss,B} = 1.32 \times 10^6 R_L$ W

$\dfrac{P_{Loss,A}}{P_{Loss,B}} = 1.28$, i.e. 28% greater than case B

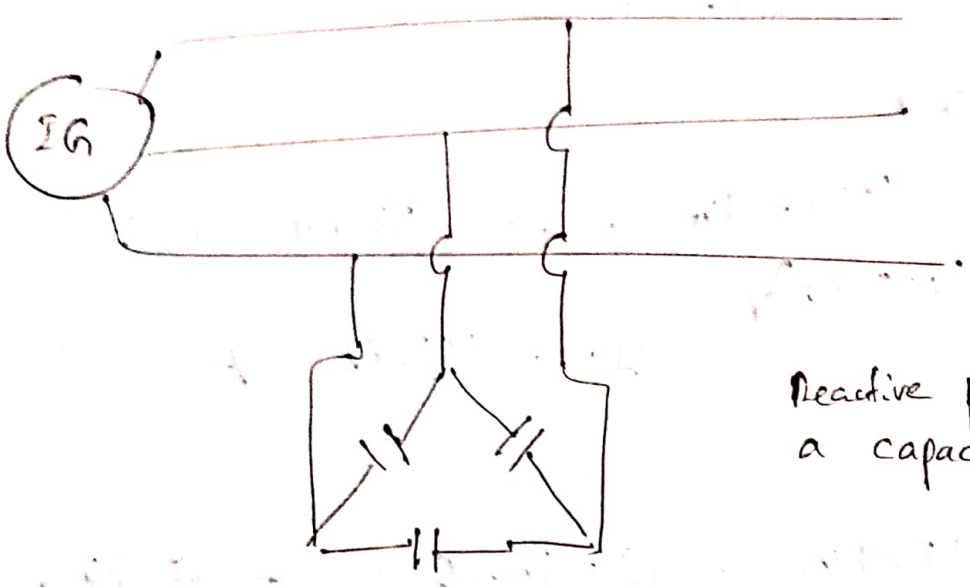
Q4

4.1)



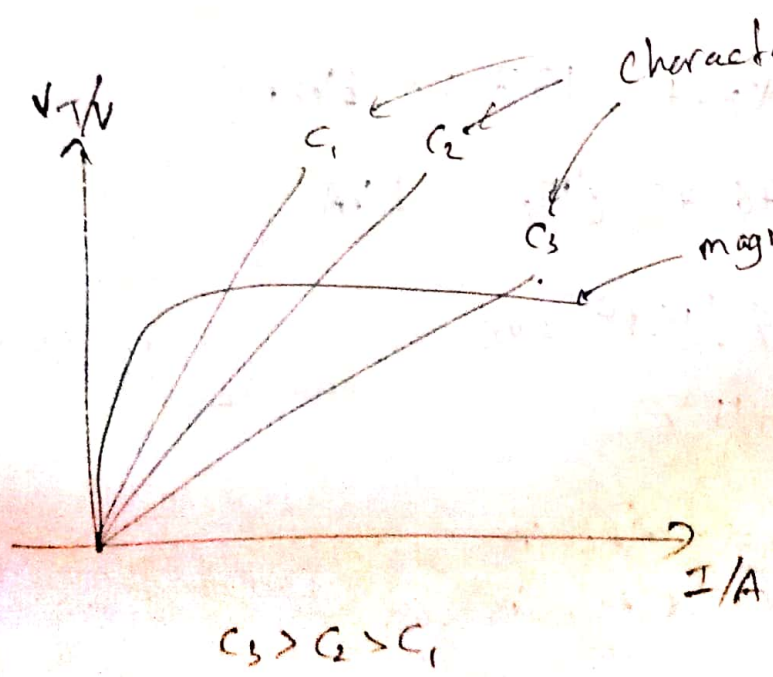
4.2

(a)



Reactive power supplied by a capacitor bank.

(b)



Intersection point is the operating point